

ALL PHOTOS BY ROBERT ERICKSON EXCEPT AS NOTED

Hand-shaping Wood For Furniture

A chairmaker discusses his tools and techniques for creating curves

BY ROBERT ERICKSON

My first wood project was a spoon for my mother when I was 22. Transforming a square block of scrap walnut into a graceful, functional ladle was a magical experience for me, and I too was transformed, hooked on the process of making things from this workable and beautiful material.

Since undergoing that transformation 36 years ago, I've spent most of my working hours designing and making furniture. My first efforts produced pieces with rounded, organic lines inspired by the style popularized by Art Espenet Carpenter in the 1960s; I was introduced to that style by Roger Somers, a charismatic carpenter from Marin County, California, whose ideas for house designs and furniture were drawn from nature and anything amoebic and curvaceous in form. As a counterpoint to Somers' style, I was simultaneous influenced by the more disciplined design work of his shop partner, Ed Stiles. Together Roger and Ed taught me that furniture design could be exciting, and Ed showed me that hand-shaping was the best way of expressing an idea in wood.

I soon fell in love with the subtle effects that sharp edges flowing into soft curves could elicit. Combining this fusion with what furnituremaker Garry Knox Bennett aptly calls "turning the corner," I've learned to dodge the stolid look that so often accompanies flat planes and right angles.

One way to emphasize a hard line is to "waste the down or back side" of certain parts of the furniture you're working on. By this I mean getting rid of a certain amount of material in order to turn right angles into acute angles. I think of this as eliminating visual bulk, and I use it on the leading edges of chair seats, on headboards, and on arms, as well as on the perimeter edge of some of my table tops. This creates a sharp, defined look and can make an object appear lighter than it otherwise would. Furnituremaker Mark Levin takes this technique to the extreme by weaving the hard line around the edge of a table top in his "Leaf Table Series" [www.marklevin.com].

In my three-and-a-half decades as a



PHOTO BY JIM BECKETT

The author's tools for shaping his chairs, such as the "Poonkinny Chair" pictured above, include electric chainsaws, angle grinders, air and electric sanders, spokeshaves, and patternmaker's files and rasps.

craftsman, I've made the usual variety of furniture, but my specialty has always been designing and making chairs, that member of the furniture family whose elements are all simultaneously on view from almost any perspective. Not only must the arms, legs, back, headboard, and seat of a chair be built to withstand weight and sometimes incredible torque, they must also work together as a cohesive, readable design.

To this end, there's a lot of different techniques I use to get the shapes I want. Basically, I've got three needs, depending on which part of the chair I'm working on. Wood seats require one set of techniques, sculptural arms and legs require another, while shaping molded joints requires a third. I'm going to describe each of these techniques in turn, starting with the way I make the templates for each part of the chair.

TEMPLATES

A certain level of symmetry is important, and being able to reproduce a form is equally so. While the sculptor strives to make a single and unique beautiful form, the furnituremaker who wants to make the same piece more than once must be able to replicate shapes. In some cases, the shapes must also be mirror images of one another, as in the left and right arms of a chair. That's where the template comes in. It gets you off on the right foot, helping you duplicate shapes in one or two planes.

My starting point with any new design is to sketch, sketch, sketch, until I like what I see on paper, even if it's only a suggestion of the final piece. While some furnituremakers move from the drawing board to computer renderings or constructing models in cardboard or balsa wood, I like to go to a full-size mock-up, which I typically make in pine. The first step I take in my workshop, then, is to make a template for each element of a newly-designed chair.

I start by sketching the elements onto inexpensive scrap plywood until I get the shapes I like in the plan (top) view. Sometimes I find that it's useful to make two templates for each element: one for the plan view, one for the elevation view. After drawing, I use a bandsaw to cut out the template. Then I use a spokeshave, a hand plane or a block of wood with sandpaper to fair the curves; that is, I refine the perimeter to the lines I want.

Next, I transfer the shapes onto 1/4" Baltic birch plywood and, again, cut them out and fair the curves. I find the birch ply easier to work with when fairing a line. It also holds its shape after many uses, and it's a lot more convenient to store than a thicker piece of wood where space is limited. The walls of my shop are crowded with templates hanging three and four deep on long nails, like bags of candy in a grocery store. Over the years, I've saved a lot of money by starting with scrap wood before cutting into the expensive stuff.

SCULPTED ARMS AND LEGS

Much of the impact of the shaping comes from the arris, the sharp edge formed where two surfaces meet at an angle. This line creates an interplay between hard and soft edges, while defining form and lending crispness to a piece.

Once I've got my templates for a chair's arms and legs, I use a three-step process to shape these pieces:

1. I bandsaw the plan and elevation views of the piece of wood that I'm shaping.

2. I create and refine broad chamfers along the top and bottom edges of the piece by removing material with a bandsaw, rasp, and/or small belt-sander.

3. I use a belt-sander to round some sur-














A series of templates, often a top and side profile for each part, are used to transfer the shapes onto the stock, which are roughed out on the bandsaw. Using his finger as a guide, the author scribes lines for cutting chamfers along each edge. Once all the lines are laid out, the material can be bandsawn, filed, or sanded to create a set of parts with four chamfers, two on top and two on the bottom. This is the first phase of "turning the corner"—going from a flat surface to a sculpted shape.

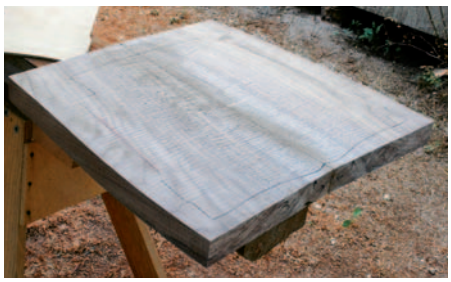
faces, and I leave arrises where I want to create definition.

After bandsawing in two directions, I create four chamfers on the piece—two on the top, two on the bottom. This is what allows me to "turn the corner"—break from a flat surface into a more sculpted, organic form. There's no formula for where to put these chamfers. In fact, one of the most enjoyable parts of designing a new piece is deciding just where they go. If you're a beginner, it helps to pencil two guidelines onto the wood for each chamfer, one along the top surface, the other along the side. An easy way to scribe a line that is evenly distanced from an edge is to use your ring finger as a reference.

Then, using a bandsaw (alternatively, a safer way is to use a patternmaker's file or even a small belt-sander), remove the wood between the lines, leaving a flat surface. Work flat to the lines. This is important. You need to make an accurate flat surface in order to perfect the rounded shapes that will come from them. You can put any angle on the chamfers that you want; you can even change the angle as you move along the edge of the piece of wood, an effect akin to the twisting, banked track of a roller coaster. Your pencil guidelines can show this. Allowing the lines to move can be an exciting design element.

Once you've got the four chamfers, you have a block of wood with eight surfaces. Using a spokeshave, a rasp, a small belt-sander or a block of wood with sandpaper, refine the surfaces you have created.

This technique of working with refined chamfers helps keep symmetry on repeatable forms. It can also be used to make round, continuous shapes. If I want a round nose on a chair arm, say, I go through all of the steps above and then create additional chamfers above and below each of the four already in place. This creates finer lines and sixteen surfaces or facets















that are easier to roll together with a sander. Good shapes come from clean, clear facets.

One of my favorite tools for these jobs is the Makita 9031 1-1/8" belt-sander. I use it in three distinct ways:

1. The "nose" is used for plowing out an inside curve.

2. The "shoe" is used for making flat chamfers.

3. The "neck" is used for the last rounding-over stage where a flat surface is not wanted.

I use this tool in many situations where other woodworkers would use a spokeshave or block plane. It has less tendency to tear out in figured woods. I also use it for fairing out lines.

CHAIR SEATS

Wooden chair seats feel better if they're not flat. It's what I refer to as the "John Deere principle": if you're going to be sitting on a metal tractor seat bumping over a rocky field all day long, you'll be a lot more comfortable if the seat is scooped out. The scooping takes some weight off your sitting bones, or whatever they're called—those two hard protuberances in your rear end that make an upholstered seat so attractive.

After gluing up the blank for the seat, the plan view is traced onto the top. Once that is cut to shape, a half-pattern is used to lay out the scooped seat area. Another pattern piece defines the front elevation. The author uses an electric chainsaw to rough out the scoop, beginning with front-to-back kerfs and then sweeping out material at right angles to that. The scoop is refined, first with a grinder, then with an orbital sander. Finally, the front edge is shaped with a 1-1/8" belt-sander.

The biggest challenge I faced when I first started scooping out seats was to find an efficient way to remove a lot of material in a time-efficient manner. Fortunately for my woodworking career, I live in a forest and am dependent on chainsaws for firewood, brush-clearing, and logging. So it didn't seem odd to me that I grabbed a chainsaw one day about three decades ago to help me with a chair. Ever since then, it's been my tool of choice for the otherwise laborious chore of shaping seats.

This is how I turn a 2" piece of lumber into a graceful seat:

1. Use a template to outline the seat from the top.

2. Use a second template to outline the edge of the scoop.











Some shaping of joint areas is done prior to glue-up. Once the parts (for example, an arm and leg) are attached, the molded joint is refined using a series of sanders with different radii. At the same time, the author carefully rounds the work by a process of transforming the eight facets (the rough shape with four chamfers) into sixteen. As the shape becomes more and more rounded and refined, arrises (sharp edges where the rounded surfaces meet) are created to give the final definition to the chair parts.

3. Chainsaw kerfs in the seat, front-to-back, to a certain depth.

4. Use the chainsaw to sweep out the kerfed material.

5. Use a 36-grit pad on a body grinder to smooth out the chainsawn seat scoop.

6. Refine the shape and smooth the surfaces with a random orbital sander.

MOLDED JOINTS AND INSIDE CURVES

Molded joints in the Danish Modern or "Sam Maloof" style frequently call for inside curves to be shaped and smoothed. A molded joint can be put together in many ways: with screws, dowels, or a mortise and tenon, for example. The important element, common to all approaches, is to leave extra material so that you will have something to shape—you start with a blocky, oversized joint.

I don't own a lathe, so I use a bandsaw and an array of sanders to cut and shape all of the parts for molded joints. Some joints are more easily shaped and sanded before the glue-up. Before joining the two elements together, I remove as much of the extra material as possible with the bandsaw. At this point I like to do a dry assembly and then use the "nose" of the Makita 9031 belt-sander to rough out the inside curve. Once I have all the chair's joints coarsely shaped, I'll glue them up. After that, it's a matter of refining all the curves and then doing the smoothing, which is my term for sanding to 400 grit.

So much of the elegance of well-shaped molded joints comes from having the right size radius for creating the pullout. For this reason I have different size sanders to help make tight radius curves. The Makita 9031 has a 1/2" radius in its nose; a Singley cylinder sander has a 3/8" radius; the Nitto Kohki pneumatic has a 1/4" radius; and the Bosch 1278VS has a 3/16" radius. Generally, the Makita removes material more quickly and the tool itself is very durable. The Bosch doesn't seem to be quite as sturdily built and doesn't hold up as well in the heavy day-to-day use that we put it to, but it makes a tighter radius and creates a smoother transition. If you "waste" as much as you can with the Makita and clean up with the Bosch, it is a nice combination. I usually follow the Makita-Bosch sequence with the Singley cylinder sander to smooth out this inside curved and molded joint. (I like to use the Nitto Kohki pneumatic sander for fitting joints where I chalk one part and rub it against the other. The Nitto Kohki is great for removing the marked material.) Cleaning up and smoothing out the joints and shapes is best done with a 5" random orbital pad sander with a flexible edge.

A friend of mine, a Pulitzer-prize winning poet, likes to tell his students that the process of writing is a craft. He finds that this helps them appreciate the kind of work that's required to produce good prose. I think the metaphor works the other way, too: mastering the craft of shaping wood is like learning a language. Just as it takes time and practice to understand the subtleties of the written word, so too it takes time and practice to understand how routing a corner, carving a finial, or gouging a chair seat affects the definition of a piece of furniture. With experimentation and practice, a woodworker becomes accomplished and develops a personal style by learning how to shape and where to shape. Sam Maloof once remarked, "It's one thing to make that hard line, and it's another to make it mean something." I couldn't agree more.

Robert Erickson is a chairmaker; he lives in Nevada City, California.